SCHULICH School of Engineering

COURSE OUTLINE
FALL 2026

Approval date	Aug 27, 2026
Approved by	DTW
Approver Department	ESE

ENEL 585 – Introduction to Power Electronics

1. Course Description

Commutation. Diode rectifiers. Fully controlled 3-phase rectifiers. Choppers, inverters, ac controllers. Single-phase switch mode converters: dc-to-dc, ac-to-dc, dc-to-ac. Circuit and state-space averaging techniques. Switching devices and magnetics.

Prerequisite(s): Electrical Engineering 465 or 469

Course Hours: 3 units; H(3-1T-2/2)

Academic Credits: 3

Calendar Reference: https://calendar.ucalgary.ca/courses/1155181

2. Timetable

Section	Days of the Week	Time	Location
LEC 01	Tu, Th	11:00-12:15	SB 105
LAB B01	Fr	8:00-9:50	ENA 13
LAB B02	Fr	8:00-9:50	ENA 13
TUT T01	Tu	10:00-10:50	SB 146

3. Course Instructors

Name: Philippe Gray
Section: 01
Email: philippe.gray@ucalgary.ca
Office: ICT 342
Phone: 210-6717

Teaching Assistants

Section	First Name	Family Name	Email
T01 (tutorial)	JT	Luterbach	jt.luterbach@ucalgary.ca
B01, B02 (laboratory)	Phuc	Luong	phuc.luong@ucalgary.ca

Laboratory technician: Mohammad Islam — mohammad.islam3@ucalgary.ca

4. Learning Outcomes

Upon successful completion of the course, you will be able to:

1. Explain the operating principles of power electronic converters — non-isolated and isolated DC–DC, single- and three-phase inverters, rectifiers, and resonant converters — together with the power semiconductor devices and magnetic components they are built from, and interpret the waveforms, models, and relationships that describe their steady-state behaviour, losses, and thermal limits. **(GA1)**

2. Analyze converter operation by identifying the governing variables, assumptions, and constraints, and by applying appropriate steady-state models — volt-second and charge balance, harmonic and phasor methods, commutation analysis — to determine converter performance against a specification. **(GA2)**

3. Derive averaged small-signal models of switching converters and evaluate closed-loop stability and dynamic performance from them, reaching substantiated conclusions about crossover frequency, phase margin, and the limits of the model used. **(GA2)**

4. Execute converter experiments on live power hardware safely and systematically, applying required electrical-safety practice and protection measures to obtain reliable measurements without damage to equipment or personnel. **(GA3)**

5. Interpret experimental data to validate predicted converter and control-loop performance, and account for discrepancies among measurement, simulation, and theory, including the uncertainty and limitations of each. **(GA3)**

6. Define and refine a converter control design to a stated specification: establish the objectives, functional requirements, and constraints — including hardware ratings, safety limits, switching frequency, and filter resonance — generate and justify candidate solutions, and refine them iteratively using analysis, simulation, and bench results. **(GA4)**

7. Select, apply, and integrate industry-standard power electronics tools — circuit simulation, real-time hardware-in-the-loop emulation, automatic code generation to a microcontroller target, and laboratory instrumentation — and evaluate the validity and limitations of their outputs. **(GA5)**

8. Contribute reliably to a project team by completing assigned work on schedule, coordinating with teammates, and demonstrating accountability for agreed design and experimental milestones. **(GA6)**

9. Communicate converter design and experimental results in clear written technical reports, and defend design decisions orally in design reviews and demonstrations, responding effectively to technical questions. **(GA7)**

10. Demonstrate professional conduct in the laboratory by acting with accountability, integrity, and respect, meeting commitments, complying with stated safety requirements, and responding appropriately to feedback. **(GA8)**

Graduate attributes

The bracketed references (e.g., **GA 1**) identify the Canadian Engineering Accreditation Board (CEAB) Graduate Attributes that are developed and assessed through each course learning outcome. The Graduate Attributes listed below describe the knowledge, skills, and professional competencies expected of graduates from accredited Canadian engineering programs. Together, they help ensure that engineering graduates are prepared for professional practice.

GA1. A knowledge base for engineering
GA2. Problem Analysis
GA3. Investigation
GA4. Design
GA5. Use of engineering tools
GA6. Individual and team work
GA7. Communication skills
GA8. Professionalism
GA9. Impact of engineering on society/environment
GA10. Ethics and equity
GA11. Economics and project management
GA12. Life-long learning

5. Course Evaluation

Assessment Components

The following assessments will be held in this course:

- Quizzes (×4)
- Laboratory: orientation, plus three laboratories
- Final project: a) proposal; b) design review; c) final demonstration; d) final report

Quizzes. Written in person during the scheduled class period, 50 minutes. Sep 29, Oct 13 and Nov 17 in the tutorial slot; Dec 3 in the lecture slot. Permitted: one double-sided handwritten aid sheet and one sanctioned Schulich School of Engineering calculator. Not permitted: notes, textbooks, phones, laptops, tablets or other electronic devices, internet access, communication with any other person, or artificial intelligence tools. The material covered by each quiz is confirmed in class and on D2L at least one week beforehand.

Laboratory. Orientation: Sep 8 (B02), Sep 15 (B01). Lab 1: Sep 11 (B02), Sep 18 (B01). Lab 2: Oct 9 (B02), Oct 2 (B01). Lab 3: Oct 23 (B02), Oct 16 (B01). Orientation is worth 2% and is participation-marked. Each laboratory is worth 6%, marked in three equal parts: an in-person laboratory quiz held at the beginning of the lab session, the report, and professionalism during the session. Laboratory quizzes are closed book; no electronic devices, notes or artificial intelligence tools may be used. Prelabs are ungraded preparation. Reports are submitted to the D2L dropbox one week after the session; late reports are penalized 20% per day and are not accepted more than three days after the deadline.

Final project. Students work in groups of three to four on an assigned converter design project that includes experimental work on laboratory hardware. Proposal due Oct 9. Design review (oral), held during bench session 1: Oct 30 for B01, Nov 6 for B02. Bench sessions run from Oct 30 to Dec 4. The final demonstration takes place during the course's registrar-scheduled final examination slot; no

final examination is written in this course. Final report due Dec 11. The proposal, design review, demonstration and report are group-marked; the demonstration also includes an individually marked question period. The project specification and the requirements for each deliverable are issued in the project handout on D2L.

Peer contribution adjustment. At the end of the project each student completes a confidential peer assessment of their group. Where the evidence supports it, an individual student's mark on the group-marked components may be adjusted by up to ±10 percentage points of the project mark. The adjustment is made by the instructor, weighing peer input against observation at the bench sessions and performance in the individual question period.

Missed assessments. Contact the instructor within 48 hours, or as soon as reasonably possible. You may choose the form of supporting documentation that best suits your situation (University Calendar, Section M.1); a medical note is not required, and the Student Declaration of Absence for In-course Assessments form is accepted on its own. There are no make-up assessments in this course. Where an absence is accepted, the weight of a missed quiz is redistributed across the remaining quizzes and the weight of a missed laboratory across the remaining laboratories. A missed project bench session cannot be weight-shifted and no make-up session will be arranged: experimental work is a required deliverable, and a group that loses time must make it up within its remaining scheduled sessions. Where an absence is not accepted, the assessment is graded zero.

Final Grade Determination

The final grade in this course will be based on the following components:

Component	Alignment with Learning Outcomes	Weight
Quizzes (4 × 10%)	1, 2, 3	40%
Laboratory (orientation 2%; three laboratories at 6%)	1, 4, 5, 7, 9, 10	20%
Final project	2, 3, 4, 5, 6, 7, 8, 9, 10	40%
	Total	**100%**

Conversion from a score out of 100 to a letter grade will be done using the conversion chart shown below. This grading scale can only be changed during the term if the grades will not be lowered.

Letter Grade	Lower Bound	Upper Bound
A+	90%	
A	85%	90%
A-	80%	85%
B+	77%	80%
B	73%	77%
B-	70%	73%
C+	67%	70%
C	63%	67%
C-	60%	63%
D+	57%	60%
D	50%	57%
F		50%

Notes:

- There is no minimum grade requirement on any individual component in order to pass the course.
- Benches may not be energized until a member of the instructional team has confirmed the group's setup and protection measures.

Course Format and Scheduling

Lectures are delivered in person and are not recorded; lecture notes are posted to D2L after each lecture. Tutorials, laboratory sessions, project bench sessions, design reviews, the quizzes and the final demonstration all take place in person. The Tuesday 10:00–12:15 block on December 1 is converted to a laboratory session for Section B02, replacing that day's tutorial and lecture.

Coursework Submission Guidelines

Laboratory reports are submitted to the D2L dropbox one week after the session. Project deliverables — the proposal and the final report — are submitted as PDF to D2L. Both are group submissions, with all members named.

Quizzes, laboratory quizzes, and the individual question period at the final demonstration are individual assessments and must be completed without assistance from other students.

6. Learning Resources

Textbook(s)

Required Textbook:
Title: Principles of Power Electronics
Author(s): Kassakian, Perreault, Verghese, Schlecht
Edition: 2nd, 2023
Publisher: Cambridge University Press

Recommended Textbooks:
Title: Fundamentals of Power Electronics
Author(s): Erickson, Maksimovic
Edition: Third Edition, 2020
Publisher: Springer Cham

Title: Power Electronics: Converters, Applications, and Design
Author(s): Mohan, Undeland, Robbins
Edition: Third Edition, 2002
Publisher: Wiley

Technology Requirements

This course uses PLECS with PLECS Coder, Typhoon HIL Control Center, and MATLAB. Student installation instructions are posted on D2L. A separate embedded-development environment is not required, as control code can be generated and deployed through PLECS Coder.

7. Course Policies and Expectations

7.1. Use of AI tools

To be specified: The policy regarding the use of AI tools will be individually specified for each assessment in this course.

The policy on the use of generative AI tools in this course depends on the assessment.

Permitted, with disclosure — project work, laboratory reports, and study. You may use generative AI tools to help you understand course material, to draft and edit written work, and to write, debug or explain code and simulation models. If you use an AI tool in preparing anything you submit for marks, include a short disclosure statement at the end of the submission naming the tool and describing what you used it for. You remain fully responsible for everything you submit, including its technical correctness.

Not permitted — quizzes. No AI tool of any kind may be used during the four term quizzes or the in-person laboratory quizzes.

Unaided by design — oral assessments. The design review and the individual question period at the final demonstration are conducted in person and assess what you personally understand.

Undisclosed use of AI in submitted work, and any use during a quiz, are treated as academic misconduct under the policy referenced in Section 9.

7.2. Use of Calculators in Examinations

You may use any calculator you wish for studying and completing lab reports. However, you must use only one of the following sanctioned Schulich School of Engineering calculators during exams: Casio FX-260S, Casio FX-300MS, Casio FX-95ES Plus, Casio FX-300ESPLUS2 (2nd Edition), TI-30XIIS, TI-30Xa, Sharp EL-531XTB-WH, and HP10S Plus.

7.3. Reappraisals

For reappraisals of graded term work or academic assessments (including final grades), please refer to the **SSE Reappraisal of Graded Term Work and Academic Assessments Policy**, which is available at:
https://schulich.ucalgary.ca/sites/default/files/teams/1/SSE%20Reappraisal%20Policy.pdf

Reappraisal requests must be based on an alleged grading error and follow the procedures and deadlines specified in the policy. For graded term work, students must first discuss their concerns with the course instructor within 10 business days of receiving or viewing the graded work. Reappraisal outcomes may result in a grade increase, decrease, or no change. Forms are available on the Engineering Student Center D2L site.

7.4. Attendance and Participation Expectations

Students are expected to attend lectures and tutorials regularly. Attendance is required at the laboratory orientation, which is participation-marked, at all three laboratory sessions, where professionalism is one third of each laboratory mark, and at all three project bench sessions, which cannot be made up by other means.

7.5. Conduct

Students are expected to demonstrate behaviour in class that promotes and maintains a positive and productive learning environment. This includes treating classmates, instructors, teaching assistants, staff, and external partners with courtesy and professionalism; engaging honestly in all academic activities; contributing constructively to discussions and team-based work; and respecting diverse perspectives.

As members of the University of Calgary community, students are expected to demonstrate conduct that is consistent with the University's Conduct and Non-Academic Misconduct policy and procedures: https://calendar.ucalgary.ca/uofcregs/university-regulations/integrity-conduct

As members of an engineering program, students are also expected to begin developing the professional behaviours and ethical judgment expected of engineers. Throughout this course, students should demonstrate integrity, accountability, respect for others, and a commitment to public safety

and responsible engineering practice. Students are encouraged to familiarize themselves with the ethical responsibilities of the engineering profession as described in the Association of Professional Engineers and Geoscientists of Alberta (APEGA) Ethical Practice guideline: https://www.apega.ca/docs/default-source/pdfs/standards-guidelines/ethical-practice.pdf

7.6. Use of Electronic Communication Devices in Class

The use of laptop and mobile devices in class is acceptable when used in a manner appropriate to the course and classroom activities. Please refrain from accessing websites and resources that may be distracting to you or for other learners during class time. Students are responsible for being aware of the University's Internet and email use policy, which can be found at https://www.ucalgary.ca/legal-services/university-policies-procedures/acceptable-use-electronic-resources-and-information-policy

7.7. Expectations for Writing

All written assignments (including, to a lesser extent, written exam responses) will be assessed at least partly on writing skills. Writing skills include not only surface correctness (grammar, punctuation, sentence structure, etc.) but also general clarity and organization. Sources used in research papers must be properly documented. If you need help with your writing, you may use the **Writing Support** (https://ucalgary.ca/student-services/student-success/writing-support) at the **Chancellor Cuthbertson Student Success Centre** (https://www.ucalgary.ca/student-services/student-success/student-success). For further information, please refer to the official online University of Calgary Calendar, Academic Regulations, E. Course Information, E.2: Writing Across the Curriculum: https://calendar.ucalgary.ca/pages/2c2d1ce47b8c4d008aec9cc3da49876e

8. Guidelines for Recording Sessions

For more information about recording classes in any learning environment see the ***Media Recording of Students in Learning Spaces policy*** here: https://www.ucalgary.ca/provost/teaching-learning/recordings-learning-environments

8.1. Media Recording for Study Purposes

Students who wish to **audio record** lectures for personal study purposes need to follow the guidelines outlined in Section E.6 of the University Calendar: https://calendar.ucalgary.ca/pages/4ba66d5bd43b422ba2e9ab3d6331d5b0

Unless the audio recording of lectures is part of a student accessibility requirement, permission must be sought by the course instructor to audio record lectures.

8.2. Media Recording for Lesson Capture

The instructor may use media recordings to capture the delivery of a lecture. These recordings are intended to be used for lecture capture only and will not be used for any other purpose. Recordings will be posted on D2L for student use and will normally be deleted at the end of term. Students are responsible for turning off their camera and/or microphone if they do not wish to be recorded.

8.3. Media Recording for Assessment of Student Learning

The instructor may use media recordings as part of the assessment of students. This may include but is not limited to classroom discussions, presentations, clinical practice, or skills testing that occur during the course. These recordings will be used for student assessment purposes only and will not be shared or used for any other purpose. The recording will be destroyed as specified by Retention Rule 2000.01 “Examinations and Student Assignments.” https://www.ucalgary.ca/provost/teaching-learning/recordings-learning-environments

8.4. Media Recording for Self-Assessment of Teaching Practices

The instructor may use media recordings as a tool for self-assessment of their teaching practices. Although the recording device will be fixed on the instructor, it is possible that student participation in the course may be inadvertently captured. The recording will be destroyed as specified by Retention Rule 90.0011 “Draft Documents & Working Materials."
https://www.ucalgary.ca/provost/teaching-learning/recordings-learning-environments

9. University of Calgary Policies and Supports

9.1. SSE Advising and Policies

All Schulich School of Engineering students have access to a D2L site titled ***Engineering Student Centre***. Students have the responsibility to familiarize themselves with the policies available on this site.

9.2. Academic Misconduct

Academic Misconduct refers to student behavior which compromises proper assessment of a student’s academic activities and includes cheating; fabrication; falsification; plagiarism; unauthorized assistance; failure to comply with an instructor’s expectations regarding conduct required of students completing academic assessments in their courses; and failure to comply with exam regulations applied by the Registrar.

For information on the University of Calgary Student Academic Misconduct Policy and Procedure and the SSE Academic Misconduct Operating Standard, please visit:
https://schulich.ucalgary.ca/current-students/undergraduate/student-resources/policies-and-procedures

Additional information is available on the Academic Integrity Website at:
https://ucalgary.ca/student-services/student-success/learning/academic-integrity

9.3. Academic Accommodation

It is the student’s responsibility to request academic accommodations according to the University policies and procedures listed below. The Student Accommodations policy is available at:
https://ucalgary.ca/student-services/access/current-students/academic-accommodations

Students needing an accommodation because of a disability, or medical concerns should communicate this need to **Student Accessibility Services (SAS)** in accordance with the ***Procedure for Accommodations for Students with Disabilities*** available at: https://www.ucalgary.ca/legal-services/sites/default/files/teams/1/Policies-Accommodation-for-Students-with-Disabilities-Procedure.pdf. For additional information on support services and accommodations for students with disabilities, visit www.ucalgary.ca/access/

Students needing an accommodation in relation to their coursework or to fulfill requirements for a degree, based on a ***Protected Ground other than Disability***, should communicate this need by submitting the ***SSE Request for Academic Accommodation Form*** to the Associate Head (Undergraduate Studies) of the department offering the course, within 10 business days prior to the class, test, exam, or assignment at issue.

9.4. Research Ethics

Students are advised that any research with human participants – including any interviewing (even with friends and family), opinion polling, or unobtrusive observation – must have the approval of the **Conjoint Faculties Research Ethics Board** (https://research.ucalgary.ca/conduct-research/ethics-compliance/human-research-ethics/conjoint-faculties-research-ethics-board-cfreb) or the **Conjoint Health Research Ethics Board** (https://research.ucalgary.ca/conduct-research/ethics-compliance/human-research-ethics/conjoint-health-research-ethics-board-chreb)

In completing course requirements, students must not undertake any human subjects research without discussing their plans with the instructor, to determine if ethics approval is required. Some courses will include assignments that involve conducting research with human participants; in these cases, the instructor will have applied for and received ethics approval for the course assignment. The instructor will discuss the ethical requirements for the assignment with the students.

For further information see Section E.5 Ethics of Human Studies of the University Calendar: https://calendar.ucalgary.ca/pages/627ed88eb4b041b7a2e8155effac3501

9.5. Instructor Intellectual Property

Course materials created by instructors (including presentations and posted notes, labs, case studies, assignments and exams) remain the intellectual property of the instructor. These materials may NOT be reproduced, redistributed or copied without the explicit consent of the instructor. The posting of course materials to third party websites such as note-sharing sites without permission is prohibited. Sharing of extracts of course materials with other students enrolled in the course at the same time may be allowed under fair dealing.

9.6. UCalgary Notice of Collection

The University of Calgary (UCalgary) respects your privacy and is committed to ensuring the privacy of all students, staff, and community members. Your personal information is being collected under the authority of section 4(c) of the ***Protection of Privacy Act*** (**POPA**). It will be collected, used and disclosed for the purposes outlined in UCalgary's Notice of Collection, Use and Disclosure of Personal

Information (https://www.ucalgary.ca/legal-services/access-information-privacy/ucalgary-notice-collection) and managed in accordance with UCalgary's Privacy Management Program (https://www.ucalgary.ca/legal-services/access-information-privacy/privacy-management-program).

Your personal information may be input into automated systems to generate content or make decisions, recommendations, or predictions.

Assignments can only be returned to the student and will be accessible only to authorized faculty and staff.

If you have questions regarding this notice or UCalgary's collection, use, or disclosure of your personal information, please contact accessandprivacy@ucalgary.ca.

9.7. Copyright Legislation

All students are required to read the University of Calgary policy on Acceptable Use of Material Protected by Copyright, available at https://www.ucalgary.ca/legal-services/university-policies-procedures/acceptable-use-material-protected-copyright-policy, and requirements of the copyright act, available at https://laws-lois.justice.gc.ca/eng/acts/C-42/index.html, to ensure they are aware of the consequences of unauthorized sharing of course materials (including instructor notes, electronic versions of textbooks etc.). Students who use material protected by copyright in violation of this policy may be disciplined under the Non-Academic Misconduct Policy https://www.ucalgary.ca/legal-services/university-policies-procedures/student-non-academic-misconduct-policy.

9.8. Sexual and Gender-Based Violence Policy

The University recognizes that all members of the University Community should be able to learn, work, teach and live in an environment where they are free from harassment, discrimination, and violence. The University of Calgary's sexual violence policy guides us in how we respond to incidents of sexual violence, including supports available to those who have experienced or witnessed sexual violence, or those who are alleged to have committed sexual violence. It provides clear response procedures and timelines, defines complex concepts, and addresses incidents that occur off campus in certain circumstances. Please see the policy available at https://www.ucalgary.ca/legal-services/university-policies-procedures/sexual-and-gender-based-violence-policy

9.9. Additional Support and Resources

For important information on the following student support and resources, please visit the Registrar's website at: https://www.ucalgary.ca/registrar/registration/course-outlines

- Wellness and Mental Health Resources
- Student Success
- Student Ombuds Office
- Student Union (SU) Information
- Graduate Students' Association (GSA) Information
- Emergency Evacuation/Assembly Points
- Safewalk

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Template revised on July 27, 2026 (HDS/KJ/AG)